

Sun spots — Solution

X22 (2022) — English translation

A1 You are presented with a table of experimental data $\omega(\varphi)$. The data was obtained based on the Doppler effect, which affects the frequency of electromagnetic radiation from the Sun recorded on Earth. Define parameters a and b .

Let's
build

a

graph

$\omega(\sin^2 \varphi)$.

We

draw

an

ap-

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from

which

we

find

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ters

a

and

b .

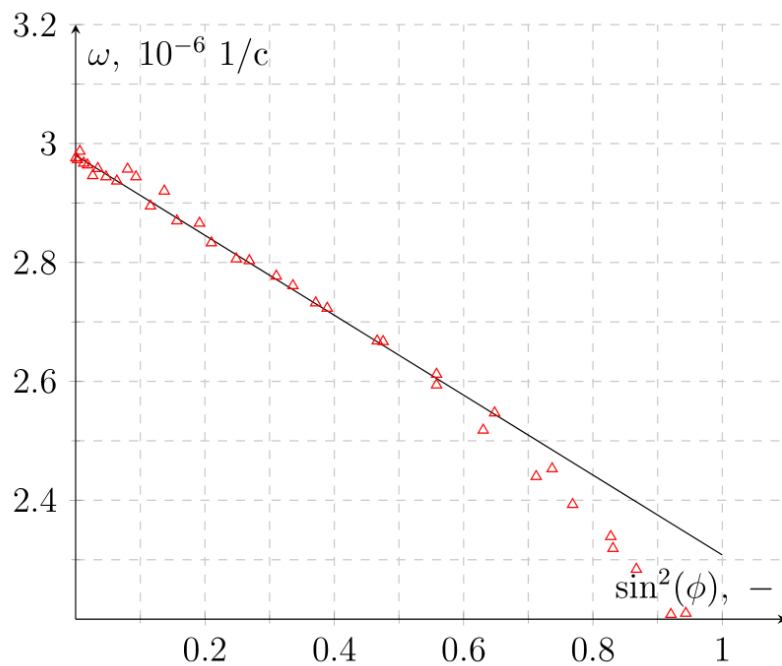
Answer:

$a =$

$(2.99 \pm$

$0.2) \cdot$

10^{-6} 1/s



A2 Find the limits of applicability of the formula given above.

From the graph we find the angle $\varphi_{\max}$ after which the points significantly deviate from the straight line.

Answer:

$\varphi_{\max} = 50^\circ$

B1 Obtain all components of the magnetic field on the surface of the Sun $B_{0,r}(\phi)$, $B_{0,\phi}(\phi)$, $B_{0,\lambda}(\phi)$. Express your answers using φ , $\mathfrak{M}$, R_0 and μ_0 .

Let
us
write
the
field
of
the
mag-
netic
dipole
at
lati-
tude
 φ :

$$\vec{B}(\varphi) = \frac{\mu_0}{4\pi} \frac{\mathfrak{M}}{R_0^3} (3\vec{n} \sin \varphi - \vec{z})$$

Projecting
this
onto
the
axes
cor-
re-
spond-
ing
to r ,
 φ
and
 λ
we
ob-
tain:

Answer:

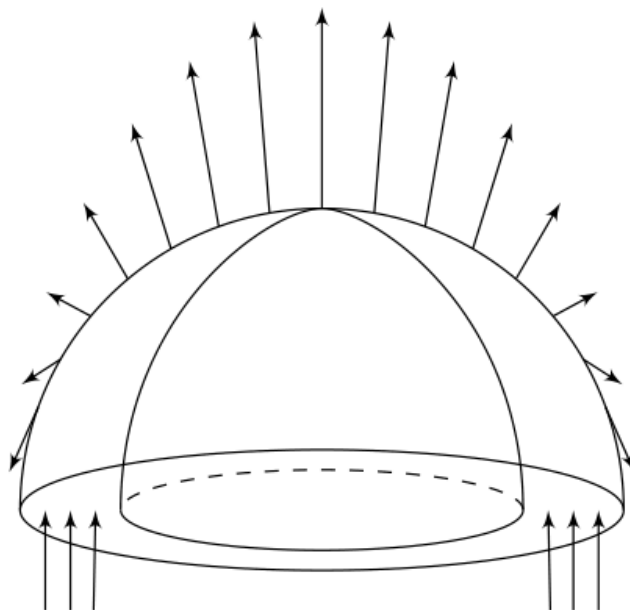
$$B_{0,r}(\varphi) = \frac{\mu_0}{4\pi} \frac{\mathfrak{M}}{R_0^3} 2 \sin \varphi$$

B2 Obtain the magnitude of the magnetic field B_e at the equator inside the Sun near its surface. Express your answer using $\mathfrak{M}$, μ_0 , R_0 , ξ .

Let's
choose
a
sur-
face
stretched
over
mag-
netic
lines.
Then
the
mag-
netic
field
flux
through
the
equa-
to-
rial
sec-
tion
in-
side
the
Sun
is
equal
to
the
flux
through
one
of
the
hemi-
spheres.

Thus
 $2\pi R_0 H_0 B_e =$
 Φ_0
means
 $B_e =$
 $\frac{\Phi_0}{2\pi \xi R_0^2} =$
53 T.

On
the
other
hand,
we
can
cal-
cu-
late



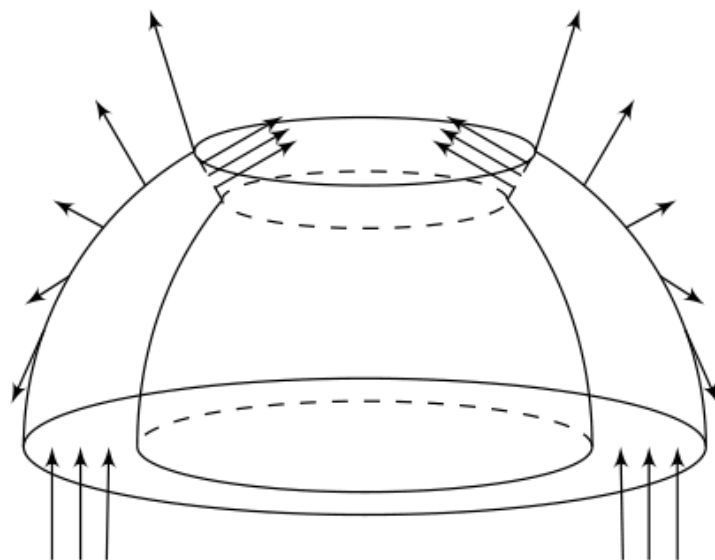
B3 Find the numerical value of B_e .

Answer: 10

$$B_e = \frac{\Phi_0}{2\pi\xi R_0^2} = 53 \text{ T}$$

B4 Obtain the magnitude of the magnetic field $B_\varphi(\varphi)$ inside the Sun at its surface as a function of latitude φ . Express your answer using φ , B_e .

Now
we
will
also
se-
lect
a
sur-
face
stretched
over
mag-
netic
lines.
Only
now
it
will
ex-
tend
from
the
equa-
tor
to
the
de-
sired
lati-
tude
 φ .
Then
to
cal-
cu-
late
 B_φ
you
need
to
find
the
flux
 $\Phi(\varphi)$
of
the
mag-
netic
field
of
the
dipole
through
the



C1 Precisely calculate 30 pairs of points (φ, λ) through which the magnetic line piercing points $\varphi = \pm 50^\circ$, $\lambda = 0$ will pass at time $t = 150$.

Let 1.70

the

mag-

netic

line

pass

through

the

points

$(\pm\varphi_0, \lambda_0)$,

where

$\varphi_0 =$

50° ,

$\lambda_0 =$

0° .

Then

its

equa-

tion

in

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lows:

$$\lambda = \omega(\varphi)t = (a - b \sin^2 \varphi)t$$

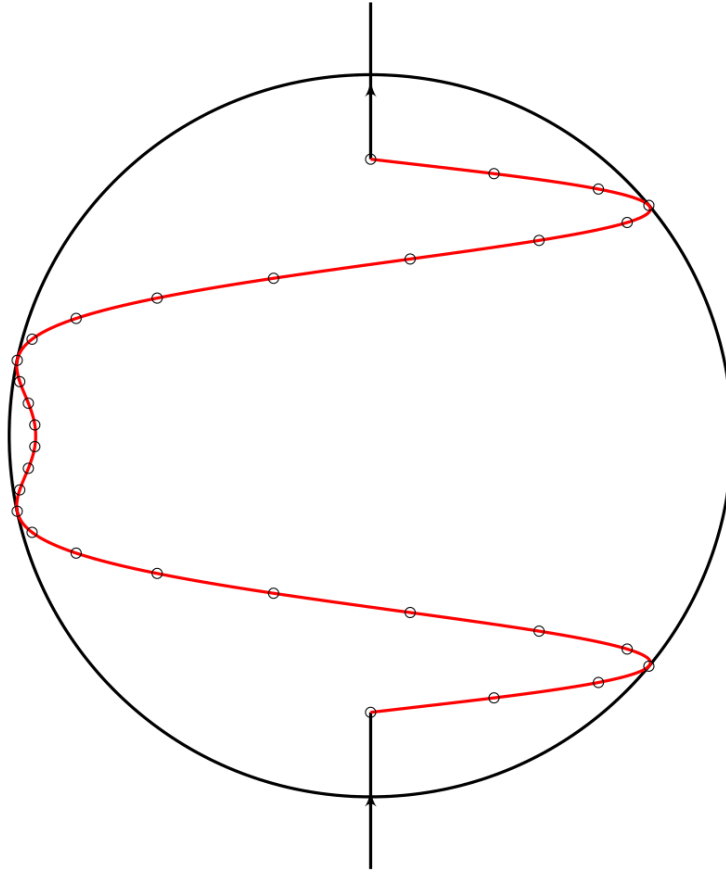
C2 Using this data, plot on your answer sheet what this line will look like for an earthly observer.

Points1.00

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in
the
plane
have
the
fol-
low-
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co-
or-
di-
nates:

$$\begin{cases} x = R \cdot \sin \lambda \cos \varphi \\ y = R \cdot \sin \varphi \end{cases}$$

Let's
draw
a
line
through
the
re-
cal-
cu-
lated
points:



C3 Find the angle $\psi(\varphi, t)$ that the magnetic lines make with the meridian. Express your answer using φ, t, a, b .

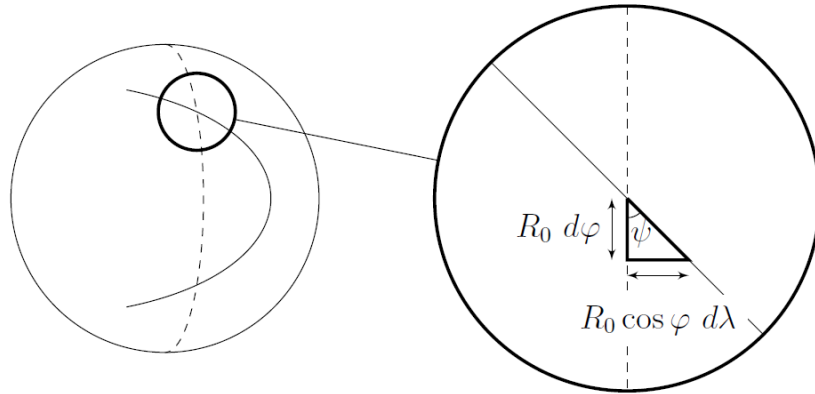
From the picture it is clear that

$$\tan \psi = \frac{d\lambda}{d\varphi} \cos \varphi = 2bt \sin \varphi \cos^2 \varphi,$$

is from

Answer:

$$\psi = \arctan (2bt \sin \varphi \cos^2 \varphi) .$$



- 4 Let the magnetic lines inside the Sun near its surface make an angle ψ with the meridian. Find the magnitude of the magnetic field B at this point. Express your answer using ψ , φ , B_e .

The magnetic field flux through the surfaces discussed in Part B should not change, so $B_\varphi = \text{const.}$ Then

$$B = \frac{B_\varphi}{\cos \psi}$$

and therefore

Answer: 20

$$\frac{B_e}{\cos \varphi \cos \psi}$$

D1 Using the $\sin \psi \approx 1$ approximation, find B .

**In
the**

**ap-
prox-
ima-
tion**

$\sin \psi \approx$

1 it

is

true

that

$\tan \psi =$

$1 / \cos \psi,$

there-

fore:

$$B = \frac{B_e \tan \psi}{\cos \varphi} = 2B_e b t \sin \varphi,$$

means

the

type

of

curve,

$\varphi(t),$

on

which

spots

ap-

pear

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the

Sun

is

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$$B = 2B_e b t \sin \varphi,$$

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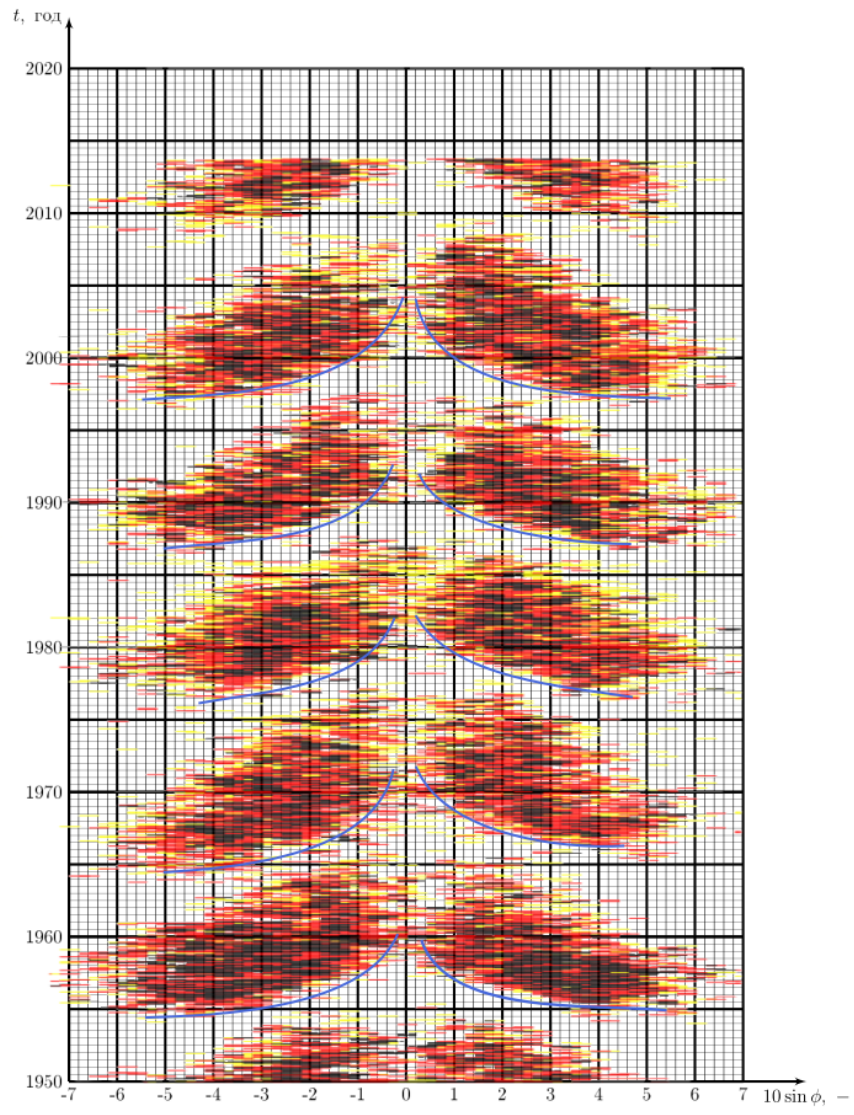
tion

is as

fol-

lows:

$$t = \frac{A}{\quad}.$$



D2 Determine the period of solar activity T_0 , i.e. the time after which the Sun's magnetic field returns to its original state.

We find the period as the time between two points with the same "phase" divided by the number of cycles:
 $T = 11$.
Next, this number must be increased by 2 times, since after one cycle the field changes direction.

Answer: 20

$$T_0 = 22 \text{ year}$$

Source: <https://pho.rs/p/506/s>